MATHEMATICS

GENERAL NOTATION, SETS AND PROBABILITY

General notation

Symbol	Meaning	Examples
$=$ $\neq$	Equal Not equal	$2+3=5$; $4-4=0$ $2\neq 3$; $-1\neq 1$
$\approx$	Approximately	$\pi\approx 3.14$ $\sqrt{2}\approx 1.41$
$<$ $>$	Less than Greater than	$2<3$; $-1<0$ $5>4$; $0>-2$
$\leq$ $\geq$	Less than or equal Greater than or equal	$2\leq 2$; $-1\leq 3$ $5\geq 4$; $0\geq 0$
$\pm$	Plus or minus	$3\pm 1=4$ or 2 $-2\pm 3=1$ or -5
$\lvert a\rvert$	Absolute value	$\lvert -5\rvert=5$; $\lvert 3\rvert=3$
$\sqrt{a}$	Square root	$\sqrt{9}=3$; $\sqrt{\frac{1}{4}}=\frac{1}{2}$

Sets and number systems

Symbol	Meaning	Examples
$\{a,b,c\}$	Set with three elements	$\{2,4,6\}$; $\{-1,0,5\}$
$\varnothing$	Empty set	$\{1\}\cap\{2\}=\varnothing$ $\{x\in\mathbb{R}: x^2=-1\}=\varnothing$
$\in$ $\notin$	Is element of Is not element of	$2\in\{1,2\}$; $-1\in\mathbb{Z}$ $3\notin\{1,2\}$; $\frac{1}{2}\notin\mathbb{Z}$
$\subseteq$	Is a subset of	$\{1\}\subseteq\{1,2\}$ $\mathbb{N}\subseteq\mathbb{Z}$
$\cup$ $\cap$	Union Intersection	$\{1\}\cup\{2\}=\{1,2\}$ $\varnothing\cup\{0\}=\{0\}$ $\{1,2\}\cap\{2\}=\{2\}$ $\{-1\}\cap\{4\}=\varnothing$
$\mathbb{N}$ $\mathbb{Z}$	Natural numbers Integers	$3\in\mathbb{N}$; $8\in\mathbb{N}$ $-2\in\mathbb{Z}$; $0\in\mathbb{Z}$
$\mathbb{Q}$ $\mathbb{R}$	Rational numbers Real numbers	$\frac{1}{2}\in\mathbb{Q}$; $-3\in\mathbb{Q}$ $\sqrt{2}\in\mathbb{R}$; $\frac{1}{4}\in\mathbb{R}$
(a,b) $[a,b]$	Open interval Closed interval	$1\notin(1,3)$; $0\in(-2,2)$ $1\in[1,3]$; $-2\in[-2,0]$

Probability and events

Example contexts
I: fair die **II**: two independent fair coin tosses.
H = heads; T = tails.
I: $A=\{1,3,5\}$, $B=\{1,2\}$.
II: $A=\{\text{HH}\}$, $B=\{\text{HH},\text{HT},\text{TH}\}$.

Symbol	Meaning	Examples
Ω	Sample space: all possible outcomes	I: $\Omega=\{1,2,3,4,5,6\}$ II: $\Omega=\{\text{HH,HT,TH,TT}\}$
ω	An individual outcome	I: $\omega=3\quad\in\Omega$ II: $\omega=\text{HT}\in\Omega$
$A=\{1,3,5\}$	Event shown as a set of outcomes	I: $A=\{1,3,5\}$: odd II: $A=\{\text{HH}\}$: two heads
$A\subseteq\Omega$	An event is a subset of the sample space	I: $\{1,3,5\}\subseteq\Omega$ II: $A\subseteq\Omega$
$P(A)$	Probability of event A	I: $P(A)=\frac{3}{6}=\frac{1}{2}$ II: $P(A)=\frac{1}{4}$
A^c	Complement of event A	I: $A^c=\{2,4,6\}$ II: $A^c=\{\text{HT},\text{TH},\text{TT}\}$
$A\cup B$	Event A or B (or both)	I: $A\cup B=\{1,2,3,5\}$ II: $A\cup B=\{\text{HH,HT,TH}\}$
$A\cap B$	Both events A and B	I: $A\cap B=\{1\}$ II: $A\cap B=\{\text{HH}\}$

Symbols acquire meaning from their mathematical context.